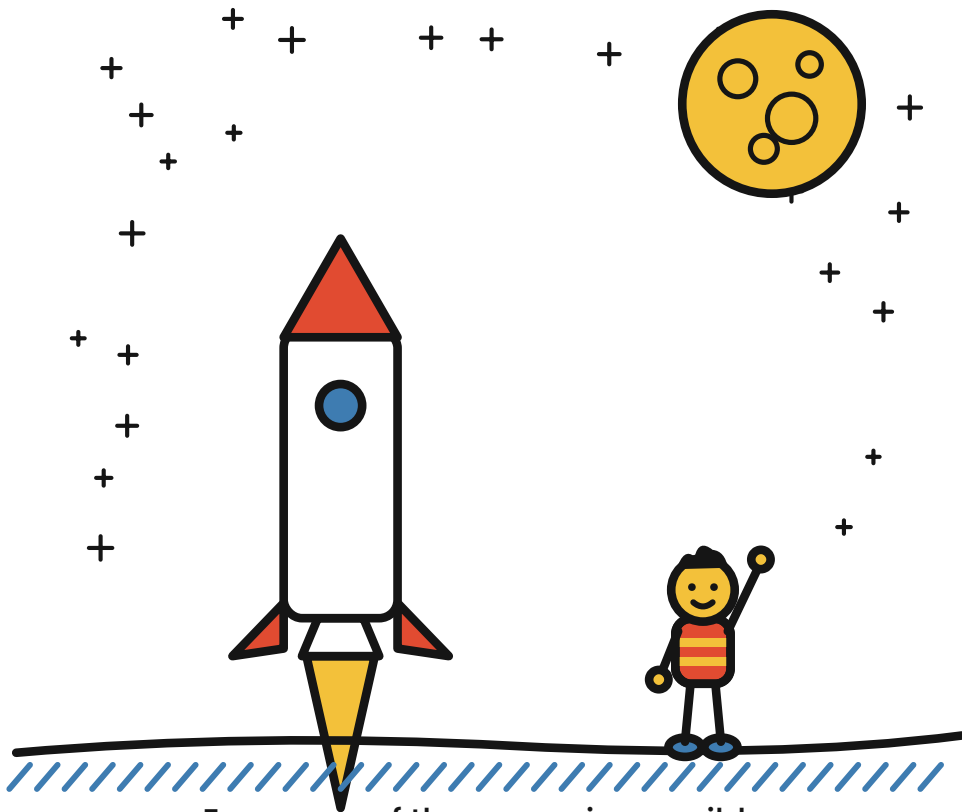


That is where we are now.

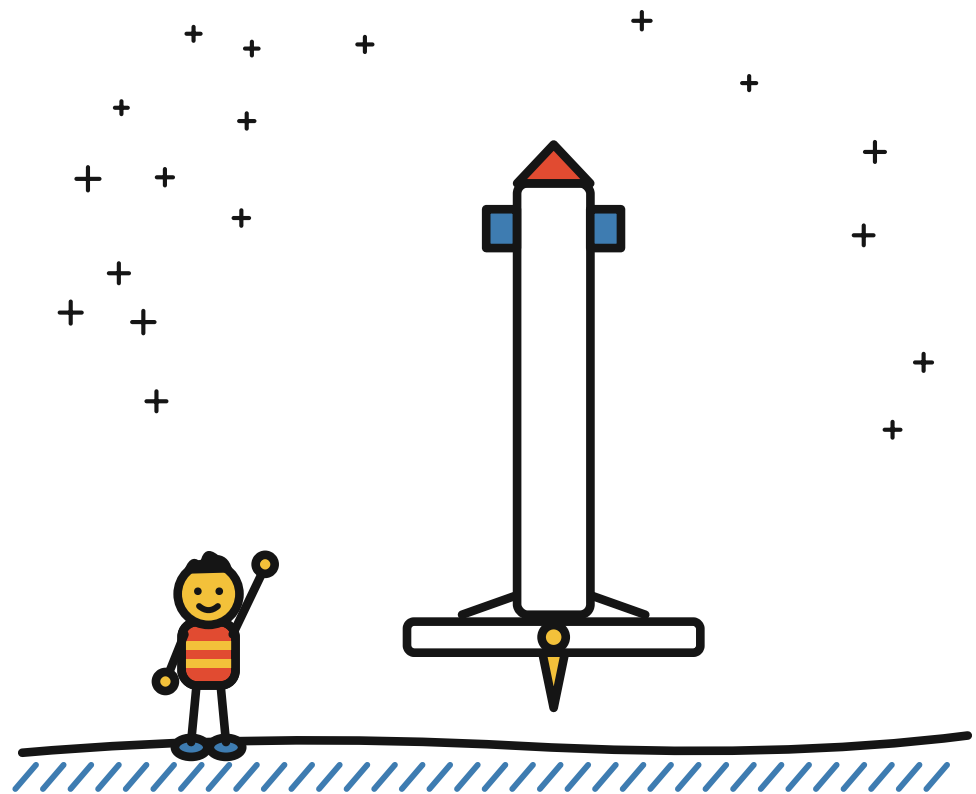
The end of the eight books · and the start of the rest



Every one of these was impossible,
until somebody kept trying.

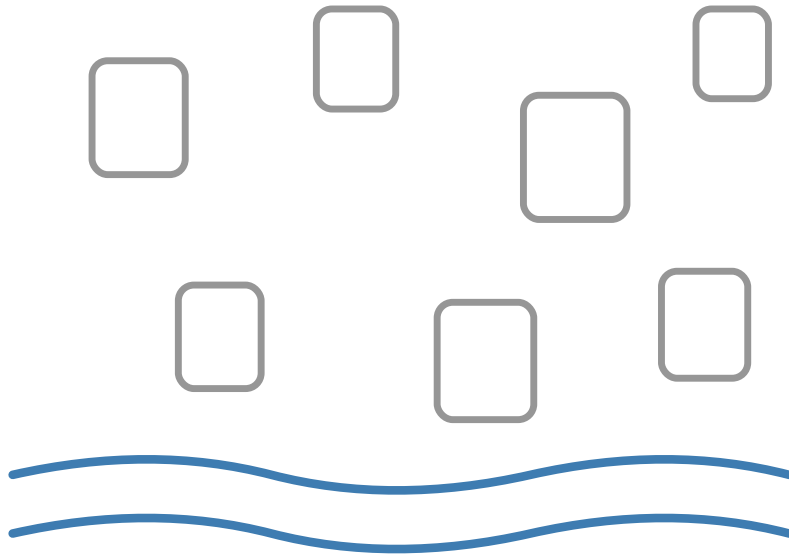
DOING IT AGAIN

Book Eight · Why we stopped throwing rockets away



Print it. Read it together. Then try, and try again.

Every rocket so far was used once.



all of it, at the bottom of the sea

Built once. Flown once. Gone.

For the grown-up.

The one idea

Reuse is not primarily about the fuel — propellant is a small fraction of the cost of a launch. It is about the hardware. Throwing away an airframe with nine engines after eight minutes of use is what made spaceflight expensive, and landing it is what changed the arithmetic.

What it actually takes

The stage separates at roughly 80 km, flips using cold-gas thrusters, burns to cancel its sideways velocity, burns again before the thick air, steers with grid fins, and lights one engine to land. It cannot hover — even at minimum throttle the thrust exceeds the empty stage's weight, so the burn must reach zero velocity exactly at the ground.

The failures were public

Grasshopper hop tests ran through 2012 and 2013; sea-landing attempts in January and April 2015 both ended in destruction on the droneship. The first successful landing came on 21 December 2015 at Cape Canaveral, the first droneship landing in April 2016, and the first re-flight of a used booster in March 2017.

What is still unsolved

Full and rapid reuse of every stage. Upper stages still burn up, and turnaround still takes far longer than an aircraft's. Vehicles aiming at this are in testing and still failing regularly — which is the honest note this series should end on, and the one worth passing on to a child.

Try this tonight.

1 Balance a broom on your palm.

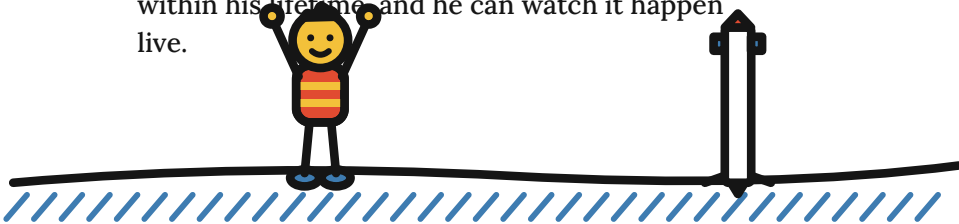
Keep it upright as long as you can. You are constantly moving your hand under it to correct. That is exactly what a landing rocket does, thousands of times a second, with engines instead of a hand.

2 Build a tower and knock it down.

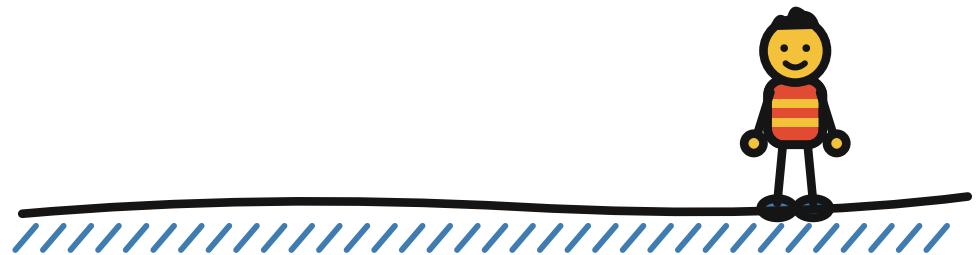
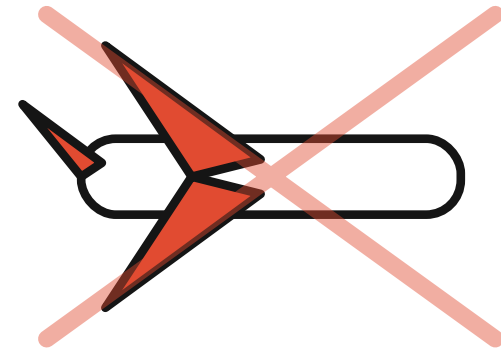
Then build it again, one block taller. Count the tries out loud. The counting is the point — nobody gets it on the first go, and everybody expects to.

3 Look up a launch to watch.

Rockets land on video nearly every week now, and the booster comes down on camera. It was impossible within his lifetime, and he can watch it happen live.

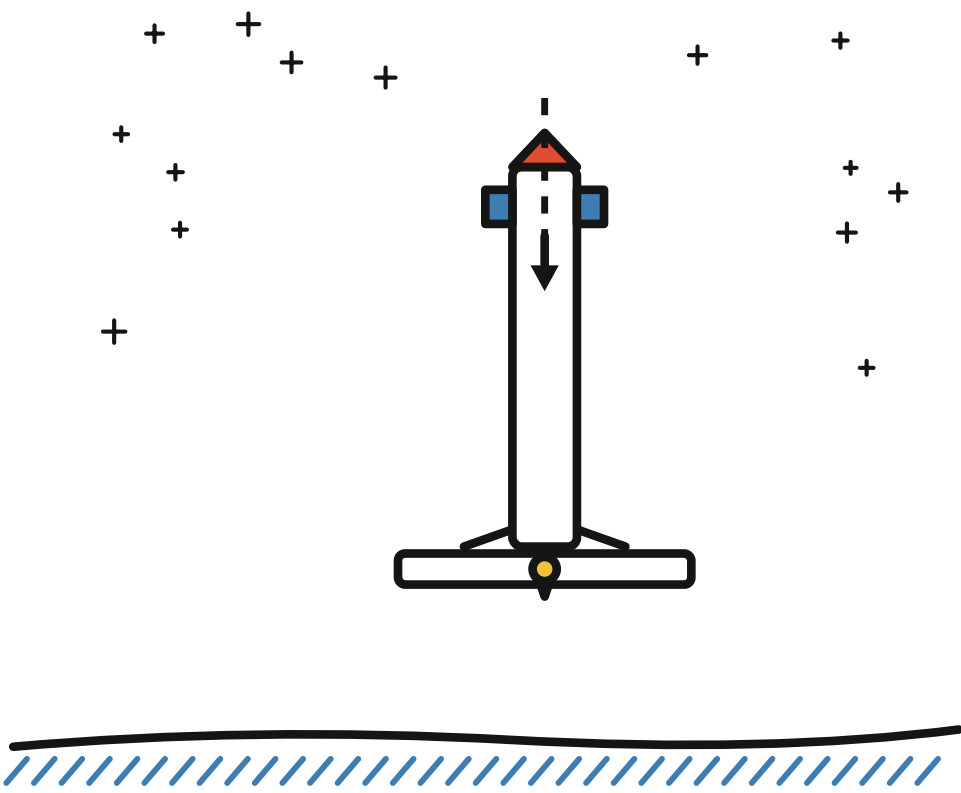


Imagine throwing away an aeroplane.



After one trip. Nobody would ever fly anywhere.

So — could it come back and land?



Not the whole thing. Just the big bottom part.

Where the story is now.

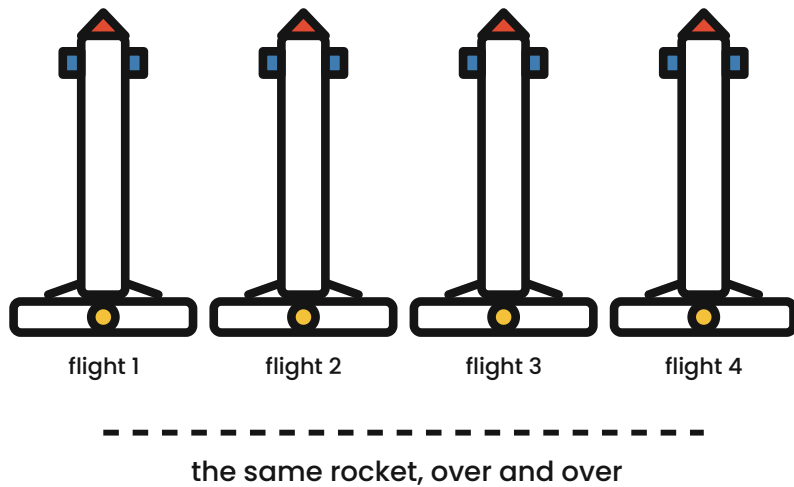
Doing It Again · figure 2 · eight books, one problem at a time

the problem	the trick	when
the air runs out	carry your own	solved, 1926
nothing to push on	throw mass backwards	solved, 1926
gravity pulls you back	go sideways fast enough	solved, 1957
fuel is too heavy	drop the empty parts	solved, 1960s
coming home is violent	a blunt shield that burns away	solved, 1961
two ships must meet	go lower to catch up	solved, 1966
rockets are thrown away	land them and fly again	solved, 2015
getting there is still costly	nobody has finished this one	still open

Every line above took someone years, and most of them took a string of public failures first.

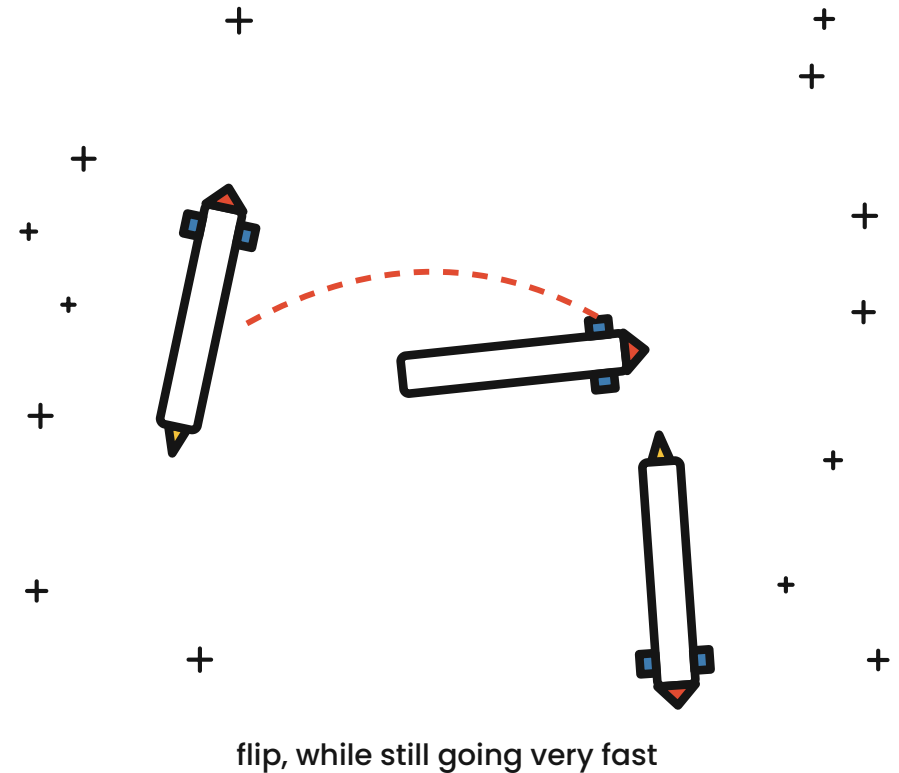
The last line is not finished. Rockets that fly like aircraft — fuelled, flown, landed, and flown again the same week — are still being built.

**Now it happens
nearly every week.**



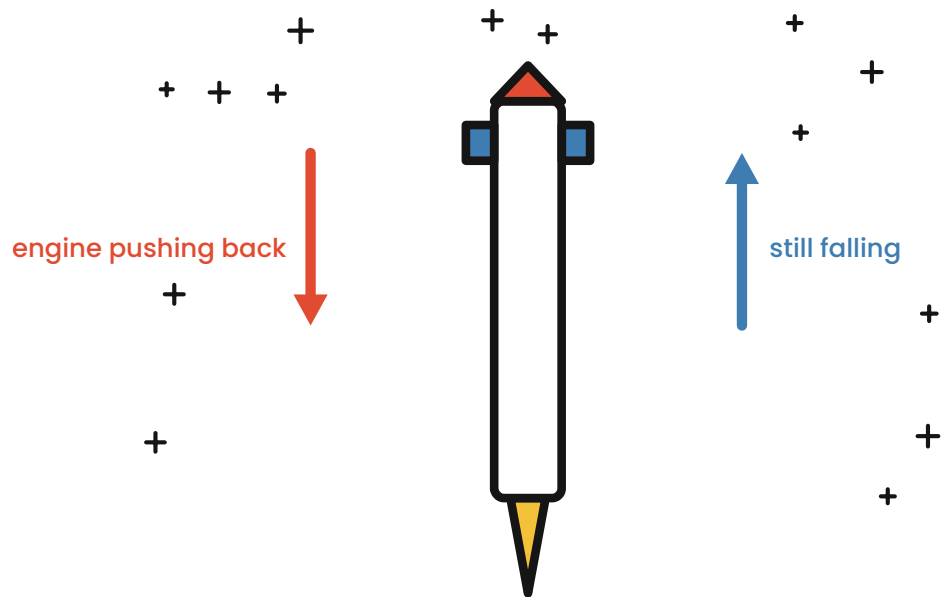
One booster has now flown more than twenty times.

**First it has
to turn around.**



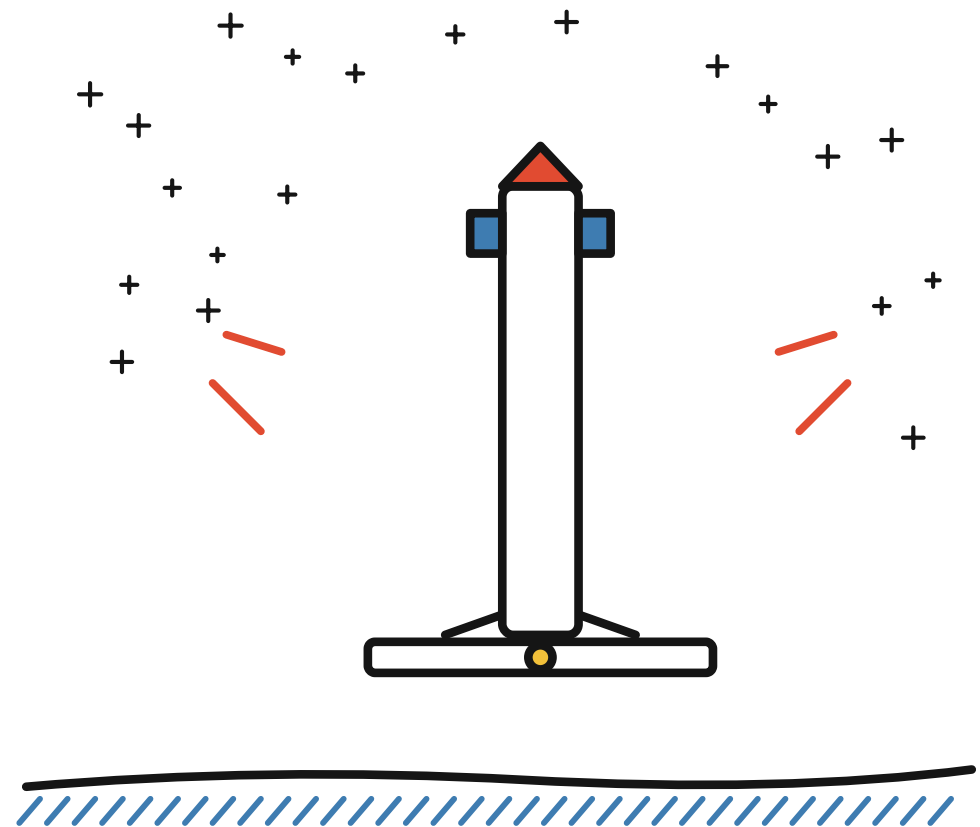
Now the engines point the way it is travelling.

Then it burns
to slow down.



It spends fuel to stop, so it can be used again.

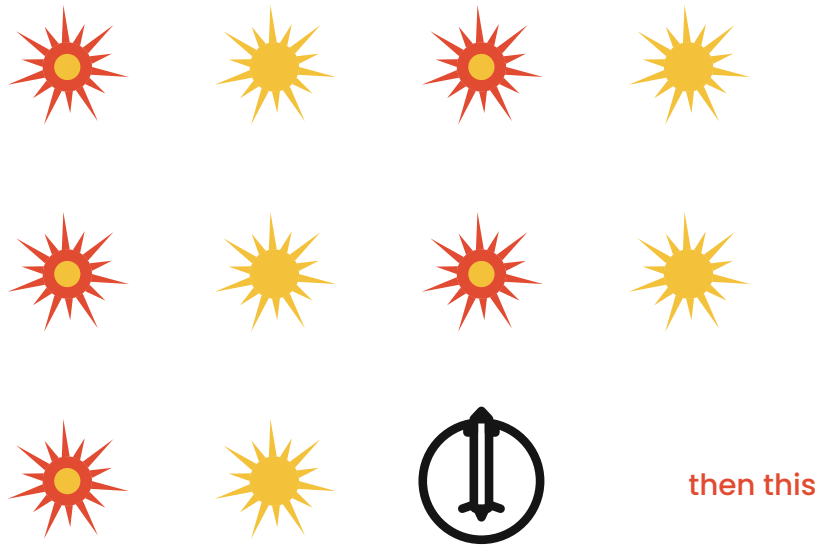
And then, one day,
it landed.



In 2015, a rocket came back,
stood up straight, and waited.

So they tried again.

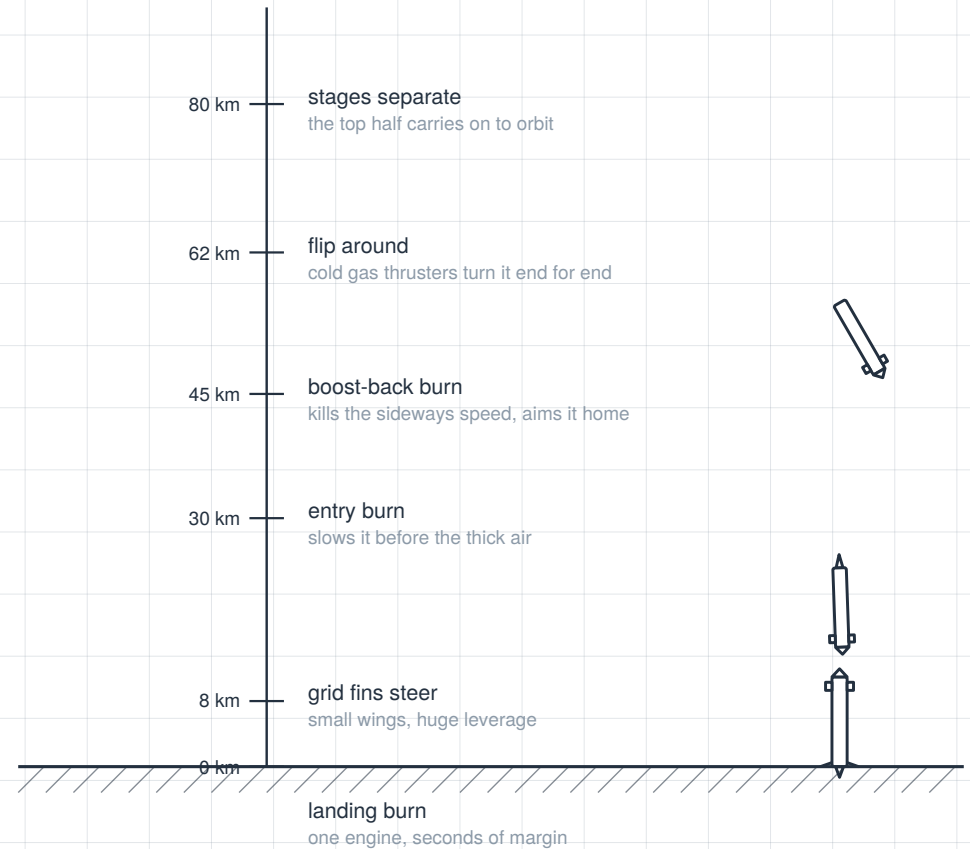
And again. And again. And again.



Every crash taught them one more thing.

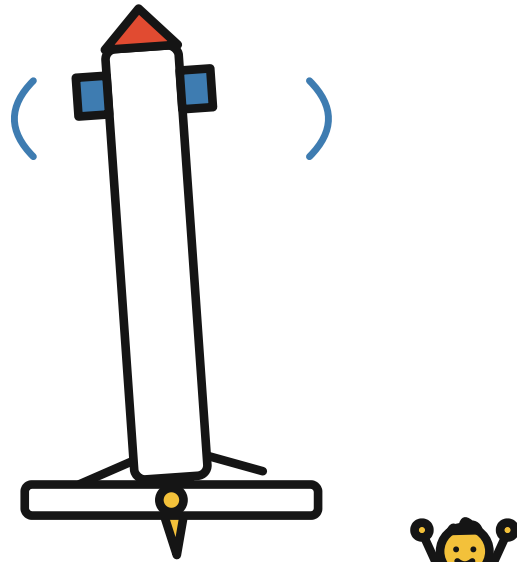
The way back down.

Doing It Again · figure 1 · a first stage returning



The fuel held back for landing is fuel that could have lifted cargo.
Reuse is worth it anyway, because the stage itself is the expensive part.

**And it must balance
on its own fire.**

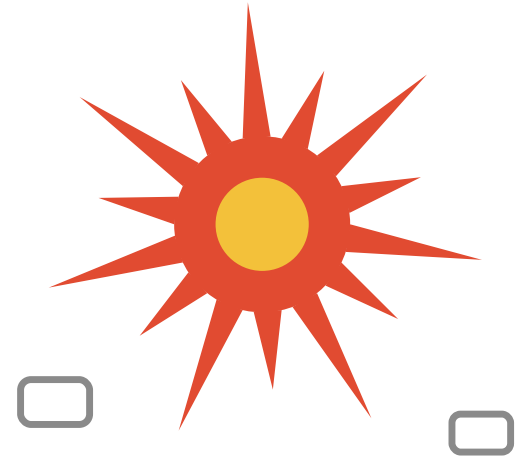


like balancing a broom on your hand

Except the broom weighs as much as a house.

8

**The first tries
did not go well.**



It fell over. It tipped up. It exploded.

9